

The type- A_2 exchange graph contradicts the proposed Weyl semidirect product

Disproof of Conjecture 00000000565

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Abstract

The coefficient-free cluster exchange graph of type A_2 is a pentagon. Its automorphism group has order 10, while the Weyl group of A_2 has order 6. Consequently its automorphism group cannot be $W(A_2) \rtimes H$ for any group H and any action. We derive the exchange graph directly from symbolic cluster mutation, and the accompanying Lean project verifies the complete counterexample.

Interpretation and the example

We use the standard unlabelled cluster exchange graph: vertices are clusters and edges are single mutations. This is the customary finite/affine-type meaning of “exchange graph” in the conjecture. The source does not give a different formal definition; in particular we do not identify this graph with a Weyl chamber graph. The type- A_2 pentagon is classical; see Fomin–Zelevinsky, Example 7.6 and Figure 3.

Work in $K = \mathbb{Q}(x, y)$, where x, y are independent indeterminates, with initial exchange matrix

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Its Cartan companion is the Cartan matrix of A_2 . In rank two each mutation replaces B by $-B$, so all exchange polynomials are $1 + z$. Thus mutating a in the cluster $\{a, b\}$ gives $\{(1 + b)/a, b\}$. Define, with subscripts modulo 5,

$$z_0 = x, \quad z_1 = y, \quad z_2 = \frac{1 + y}{x}, \quad z_3 = \frac{1 + x + y}{xy}, \quad z_4 = \frac{1 + x}{y}.$$

All five are nonzero and pairwise distinct. For example, their denominators are nonzero polynomials, and evaluation of these fractions at $(x, y) = (2, 5)$ gives $2, 5, 3, 4/5, 3/5$, respectively. Equality of two rational functions would force equality after this well-defined specialization. Specialization is used only to certify distinctness, not to replace the symbolic mutation problem by a numerical one.

All clusters and their automorphisms

Direct multiplication proves the five identities

$$z_i z_{i+2} = 1 + z_{i+1} \quad (i \in \mathbb{Z}/5\mathbb{Z}).$$

Set $C_i = \{z_i, z_{i+1}\}$. Their elements are distinct, and the five sets C_i are distinct. In C_i , mutation of z_i produces C_{i+1} , while mutation of z_{i+1} produces C_{i-1} . These are the only choices. Induction on a mutation sequence shows that every cluster reachable from C_0 is one of the C_i . Conversely, repeated forward mutation reaches all five. Hence the exchange graph is exactly the cycle

$$C_0 - C_1 - C_2 - C_3 - C_4 - C_0.$$

An automorphism of a pentagon is determined by the image of one vertex (five choices) and the image of a specified neighbor (two choices); the remaining images are forced around the cycle. All ten choices occur, via $i \mapsto b + i$ and $i \mapsto b - i$ modulo 5. Therefore

$$|\text{Aut}(\mathcal{E}_{A_2})| = 10.$$

The Weyl group obstruction

Realize A_2 using the roots $e_i - e_j$ in the plane $v_0 + v_1 + v_2 = 0$. The reflection in $\alpha = e_i - e_j$ is

$$v \mapsto v - \frac{2(v \cdot \alpha)}{\alpha \cdot \alpha} \alpha = v - (v_i - v_j)(e_i - e_j).$$

It swaps coordinates i, j . The simple reflections $s_0 = (0 \ 1)$ and $s_1 = (1 \ 2)$ generate S_3 ; its six elements are

$$1, s_0, s_1, s_0 s_1, s_1 s_0, s_0 s_1 s_0.$$

The action on the root plane is faithful: a permutation fixing all roots $e_i - e_j$ fixes their coordinate positions. Thus $W(A_2) \cong S_3$ and $|W(A_2)| = 6$.

Theorem 1. *For every group H and every action of H on $W(A_2)$,*

$$\text{Aut}(\mathcal{E}_{A_2}) \not\cong W(A_2) \rtimes H.$$

Proof. Every such semidirect product contains an injective copy of $W(A_2)$. An isomorphism would embed a group of order 6 into a group of order 10, contradicting Lagrange's theorem. $\square$

This also rules out the proposed graph-automorphism factor, whatever action is chosen. A false finite-type statement cannot acquire a valid affine extension merely by extending its scope. No claim about a separately corrected affine conjecture is needed for this disproof.

What the Lean project proves

The ambient field is the actual fraction field of the two-variable polynomial ring over $\mathbb{Q}$, using Mathlib’s `MvPolynomial` and `FractionRing` types. `clusterVar_injective` proves symbolic distinctness by cross-multiplying in this field, using the injective polynomial embedding, and only then applying polynomial evaluation. `exchange_relation` verifies the five rational identities.

`Mutates` is the two-variable A_2 rule stated above. `mutation_classification` proves that each C_i has exactly the two indicated possible successors. `Reachable` is an inductive definition allowing arbitrary finite mutation sequences; `reachable_iff` proves that its states are exactly the five listed clusters. An explicit equivalence transfers permutations of these actual states to permutations of the five cycle indices.

`exchangeAut` is the subgroup of all permutations of the reachable states preserving mutation in both directions. `exchange_aut_card` proves its order is 10. The finite enumeration involved is checked by Lean’s kernel. `root_length`, `root_pairing`, and `reflection_is_swap` verify the root reflection formula. `WeylA2` is the subgroup generated by the two coordinate swaps; `weyl_card` proves its order is 6 by exhaustive enumeration of the six words above. Finally `no_semidirect_description` excludes every second factor H , even without a finiteness assumption on H .

The formalization specializes the standard seed-mutation definition to its explicit coefficient-free A_2 rule. It does not claim to formalize a general cluster-algebra classification or a general root-system library. The connection to the source’s standard exchange-graph terminology is spelled out in the preceding proof.

Reproduction and review

Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. In `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The project is freshly rebuilt against verified official dependencies. It has no proof placeholders, custom axioms, external computational oracle, or auxiliary numerical program. The dependency audit permits only Lean’s standard axioms. Run `tectonic main.tex` to regenerate the PDF. A separate adversarial self-review was performed by the author; no independent reviewer or subagent was used.

References

1. The Last Math Competition, Conjecture 00000000565. Exact bilingual source and upstream provenance are included.
2. S. Fomin and A. Zelevinsky, *Cluster Algebras I: Foundations*, J. Amer. Math. Soc. 15 (2002), 497–529, Example 7.6 and Figure 3. Authors’ preprint.